

COMP3320/6464 High-Performance Scientific Computing Cloth Simulation

Accelerating Deep Learning with Photonics

- Computing and communicating with light**

Dr Haibo Zhang
School of Computing
Australian National University

Background

We're experiencing exceptionally high demand. Please hang tight as we work on scaling our systems. ✕

**ChatGPT is at
capacity right now**

[Get notified when we're back](#)

Write a rap about the status of ChatGPT.

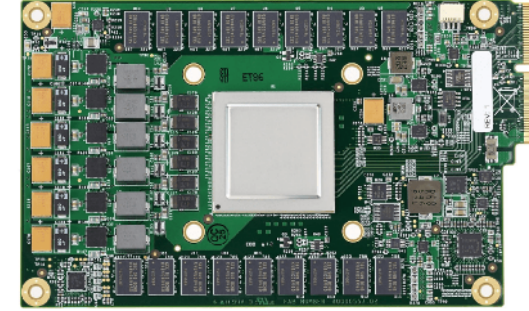
Yeah, yo, what's up ChatGPT fam
A lot of people trying to jam
But don't worry, we got your back
Just check back soon, we'll get on track

ChatGPT's the place to be
For all your AI chat needs
We're working hard to keep up the pace
So hold tight, we'll be back in this space

So don't you worry, don't be mad
We're working hard, it's not so bad
Just give us time, we'll be back
ChatGPT, the future of chat, that's a fact.

Hardware accelerators

- Google Tensor Processing Unit (TPU)
 - Specialised for **matrix and tensor operations**
 - High **memory bandwidth**
 - High-throughput processing
 - Massive parallelism
 - Reduced-precision, e.g. 16-bit floating point
- Graphics Processing Unit (GPU)
 - Massive parallelism (CUDA Cores)
 - Efficient matrix multiplication (Tensor Cores)
 - High memory bandwidth
- Field-Programmable Gate Array (FPGA)
 - Fine-grained optimization
 - Reprogrammability
 - Customization and flexibility



Google
TPUv4

144MB on-chip memory, 32GB HBM memory,
1200GB/s memory bandwidth, 275 TOPS



NVIDIA
V100

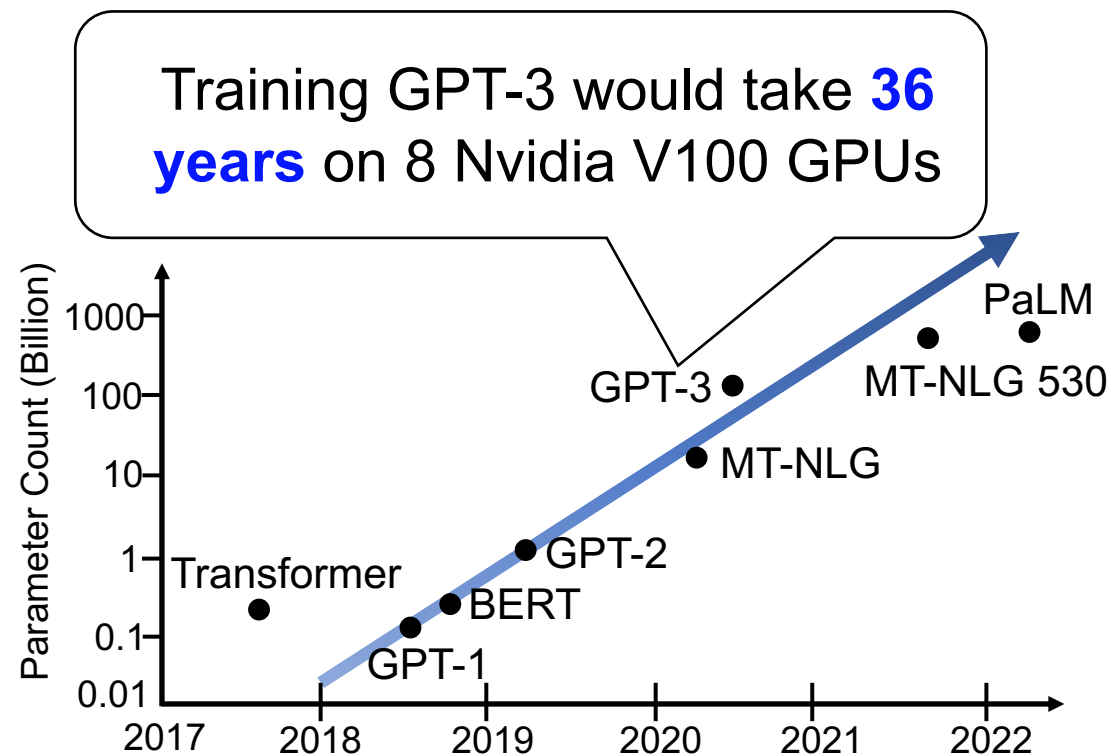
640 tensor cores; 5120 CUDA cores, 32GB HBM2
900GB/s memory bandwidth; 125 TFLOPS

Background

Current Challenges of
Deep learning

Computation-intensive
(Speed up computation)

- Models grow in size and complexity



Source: EPOCH (<https://epochai.org>)

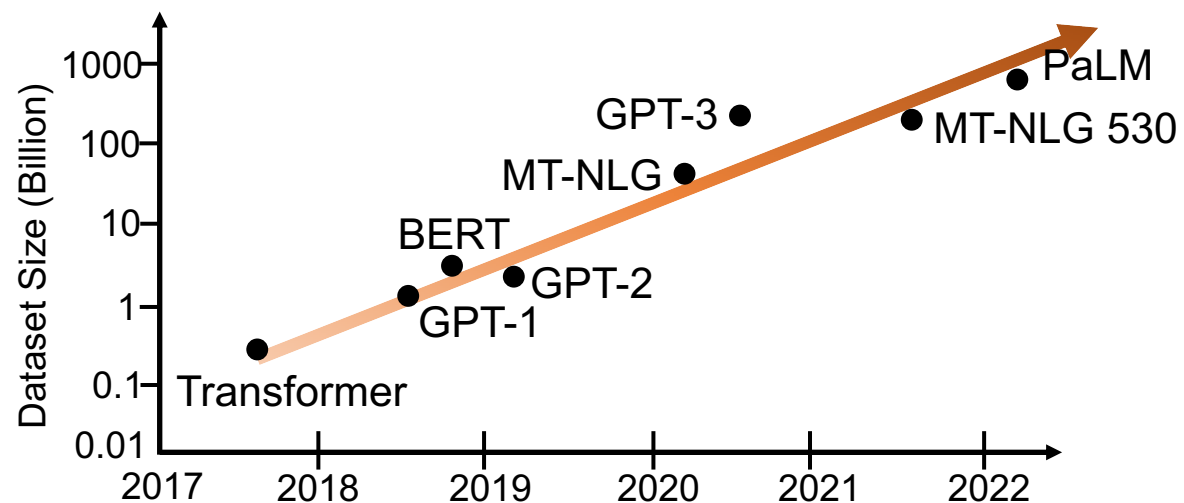
Background

Current Challenges of
Deep learning

Computation-intensive
(Speed up computation)

Memory-intensive
(Speed up data movement)

- Store training data and model parameters
- Training datasets explode in size



Source: EPOCH (<https://epochai.org>)

Background

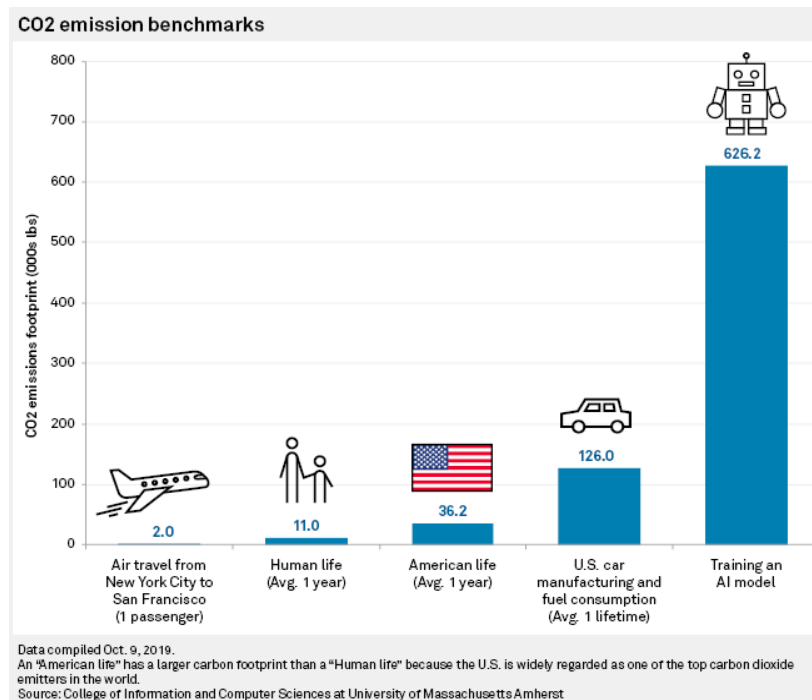
Current Challenges of
Deep learning

Computation-intensive
(Speed up computation)

Memory-intensive
(Speed up data movement)

Energy-intensive
(Need disruptive solution)

- ChatGPT energy daily - 33,000 U.S. households daily* - 8.4 tons of CO₂ per year
- Large negative environmental impact

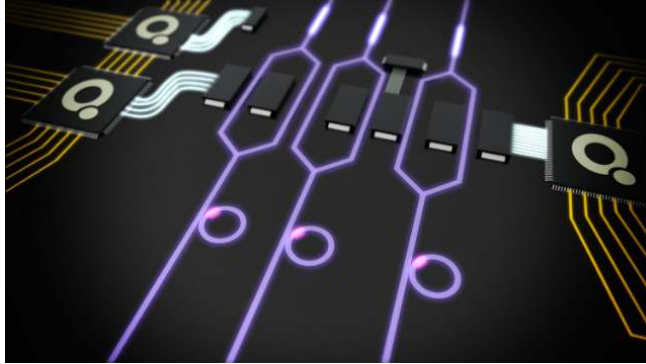


* <https://www.washington.edu/news/2023/07/27/how-much-energy-does-chatgpt-use/>

Background

Current Challenges of Deep learning	Limitations of Electronic Accelerators	Advantages of Photonics Accelerators
Computation-intensive (Speed up computation)	Limited speed/scalability (Due to Moore's law)	
Memory-intensive (Speed up data movement)	Limited memory bandwidth (Data transfer bottlenecks)	
Energy-intensive (Need disruptive solution)	High energy consumption (Large carbon footprint)	

Background



- Photonic computation at 100+ GHz frequency*
Tens of times faster than electronics
- Consuming 40 atto Joules per operation*
Hundreds of times more energy efficient than electronics
- Lightmatter's photonic chip 2021
10 times faster than the fastest GPU 90% less energy

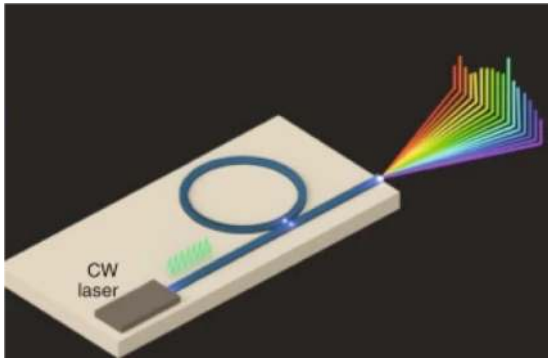
Advantages of Photonics Accelerators

High speed
(>10 Improvement*)

High bandwidth

Low energy consumption
(>100 reduction*)

Background



[Nature, 2019]



[Nature, 2020]

- > 200 parallel low-noise wavelength lines with frequency comb ([Nobel prize 2005](#))
- Optical Microcombs (RMIT/Monash/Swinburne, [\\$34.9 million ARC 2022](#), 44.2Tbps from a single light source
 - [claimed the fastest internet recorded in the world](#))

Advantages of Photonics Accelerators

High speed
(>10 Improvement*)

High bandwidth
(>100 Improvement*)

Low energy consumption
(>100 reduction*)

Background

- The Future of Deep Learning is Photonics - use light for both communication and computation.

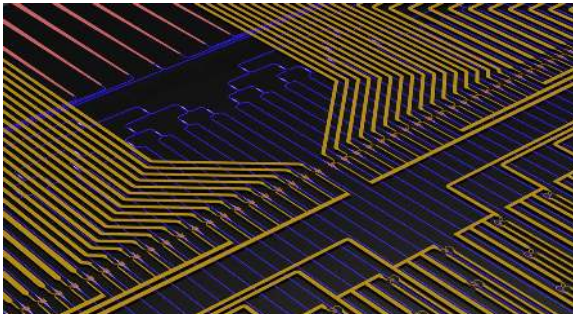
IEEE Spectrum

FEATURE COMPUTING

The Future of Deep Learning Is Photonic

Computing with light could slash the energy needs of neural networks

BY RYAN HAMERLY 29 JUN 2021



<https://spectrum.ieee.org/the-future-of-deep-learning-is-photonic>

Page 1 of 15

IEEE spectrum, 2021

MIT News

ON CAMPUS AND AROUND THE WORLD

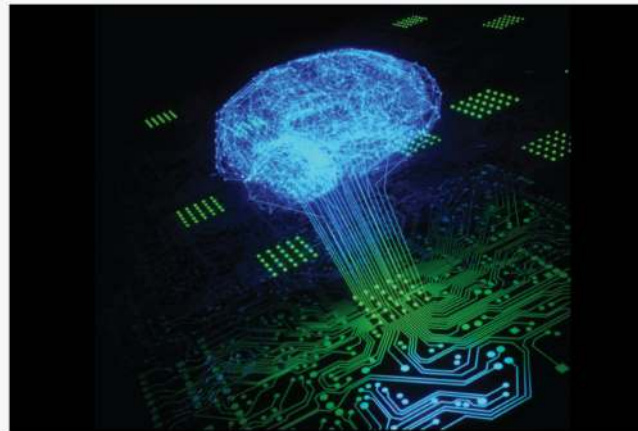
SUBSCRIBE

Machine-learning system based on light could yield more powerful, efficient large language models

MIT system demonstrates greater than 100-fold improvement in energy efficiency and a 25-fold improvement in compute density compared with current systems.

Elizabeth A. Thomson | Materials Research Laboratory

August 22, 2023



August 22, 2023, Nature

Lightmatter's photonic AI hardware is ready to shine with \$154M in new funding

Devin Coldewey @techcrunch / 10:50 AM GMT+12 • June 1, 2023

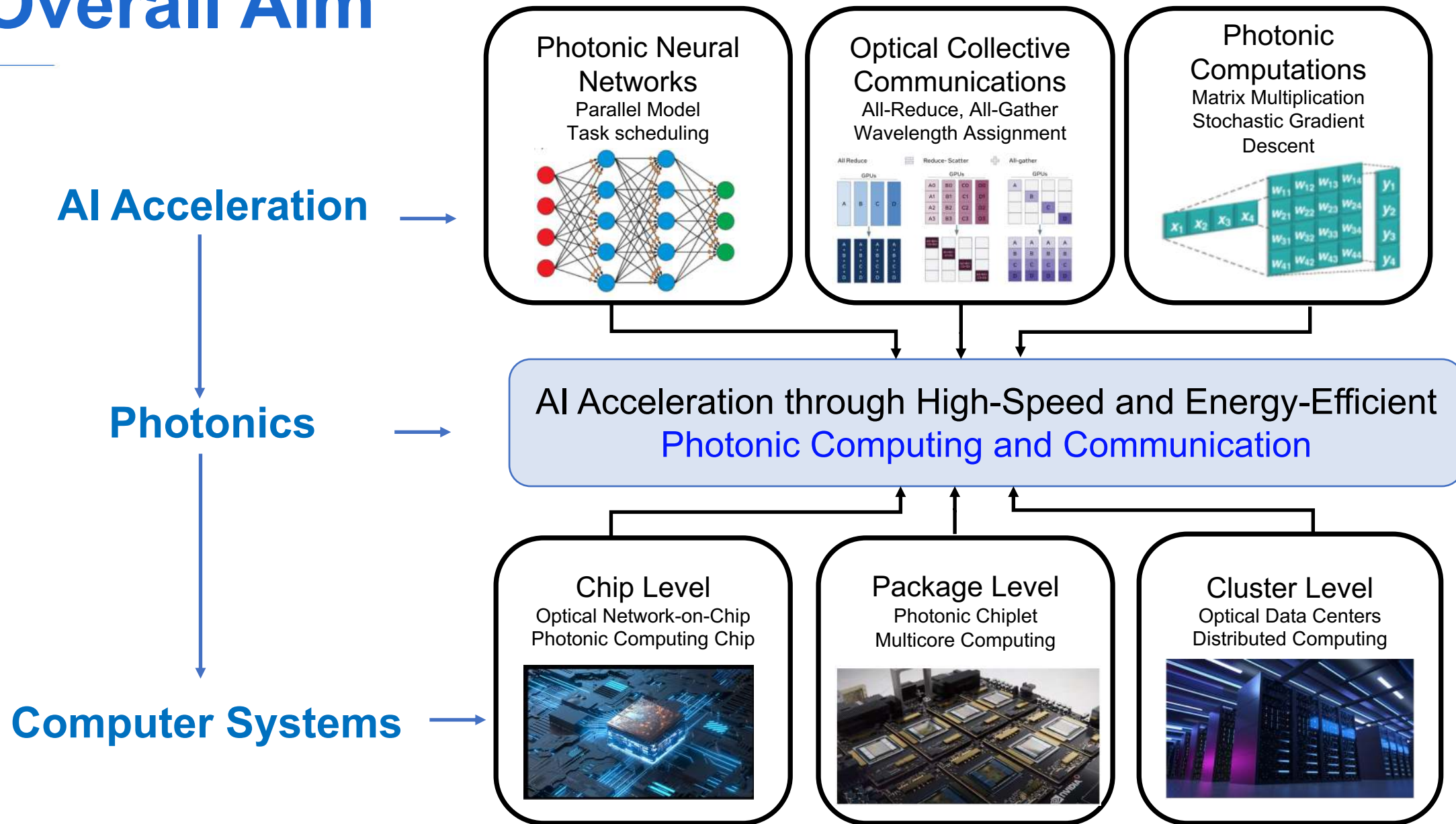
Comment



Image Credits: Lightmatter

The Future Is Bright --
Lightmatter Raises \$154M
(2023) to Deliver Photonic
Products to Customers

Overall Aim



Photonic Communication

Distributed DNN Training

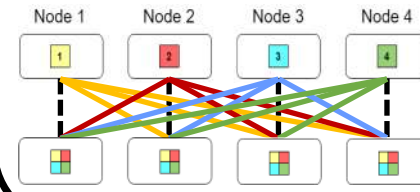
Photonic Neural
Networks

Parallel Model
Task scheduling



Optical Collective
Communications

All-Reduce, All-Gather
Wavelength Assignment



Photonic
Computations

Matrix Multiplication
Stochastic Gradient Descent



Photonic Communication

AI Acceleration through High-Speed and Energy-Efficient
Photonic Computing and Communication

Chip Level

Optical Network-on-Chip
Photonic Computing Chip



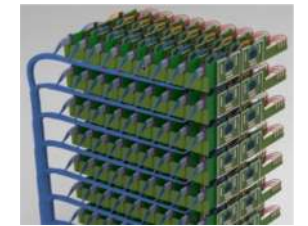
Package Level

Photonic Chiplet
Multicore Computing



Cluster Level

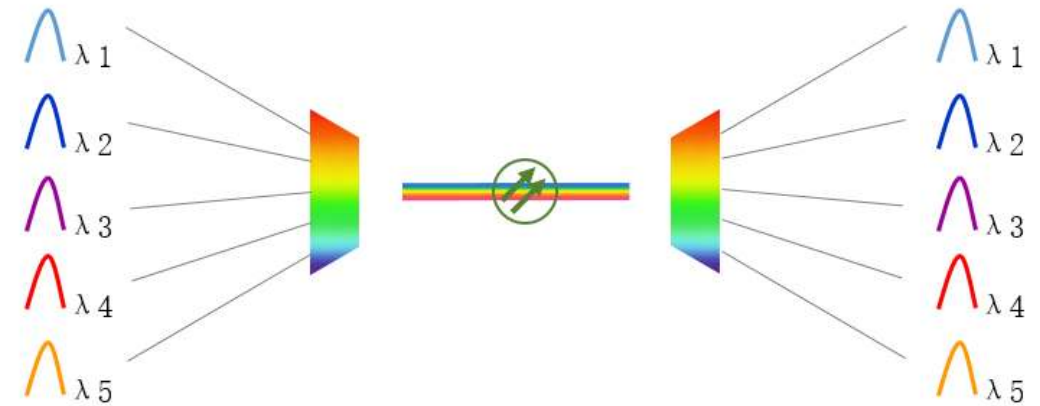
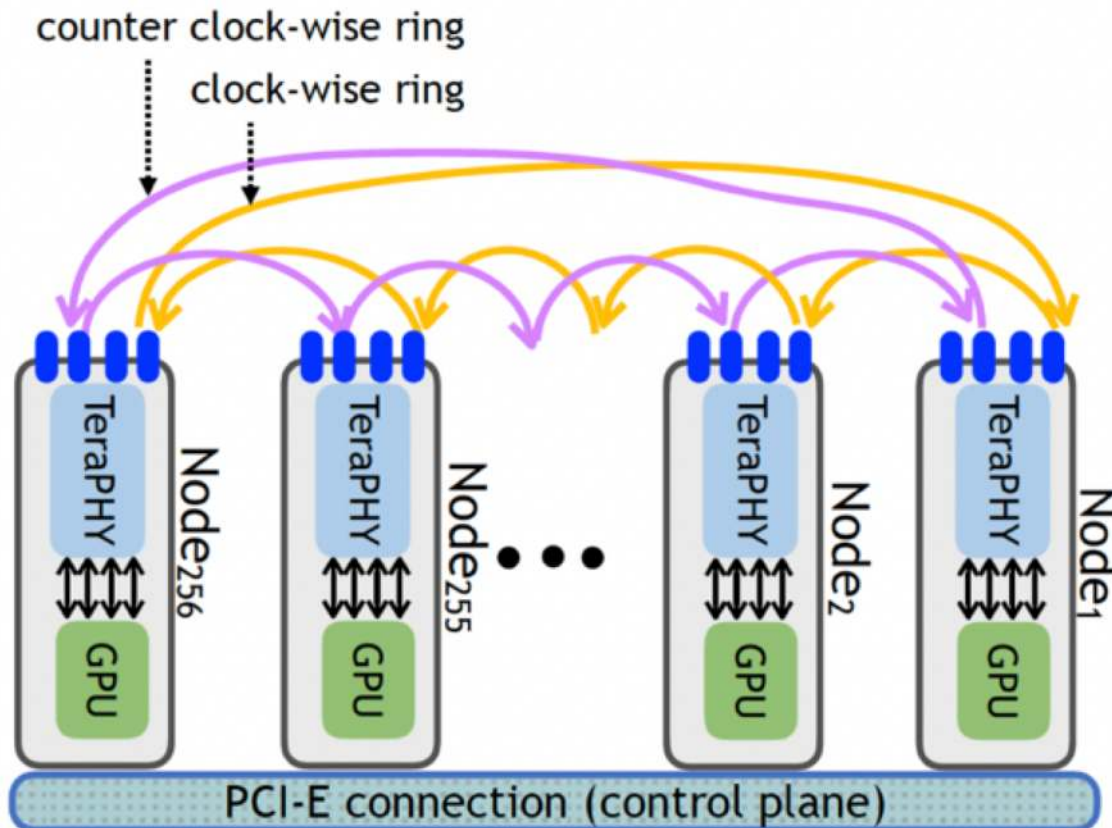
Optical Data Centers
Distributed Computing



**Inter-GPU Photonic
Interconnect Systems**

Photonic Communication

- Architecture - Inter-GPU Photonic Interconnect
 - Multiple GPUs connected using photonic links by ring topology
 - High degree of parallel communication - Wavelength Division Multiplexing (WDM)



Photonic Communication

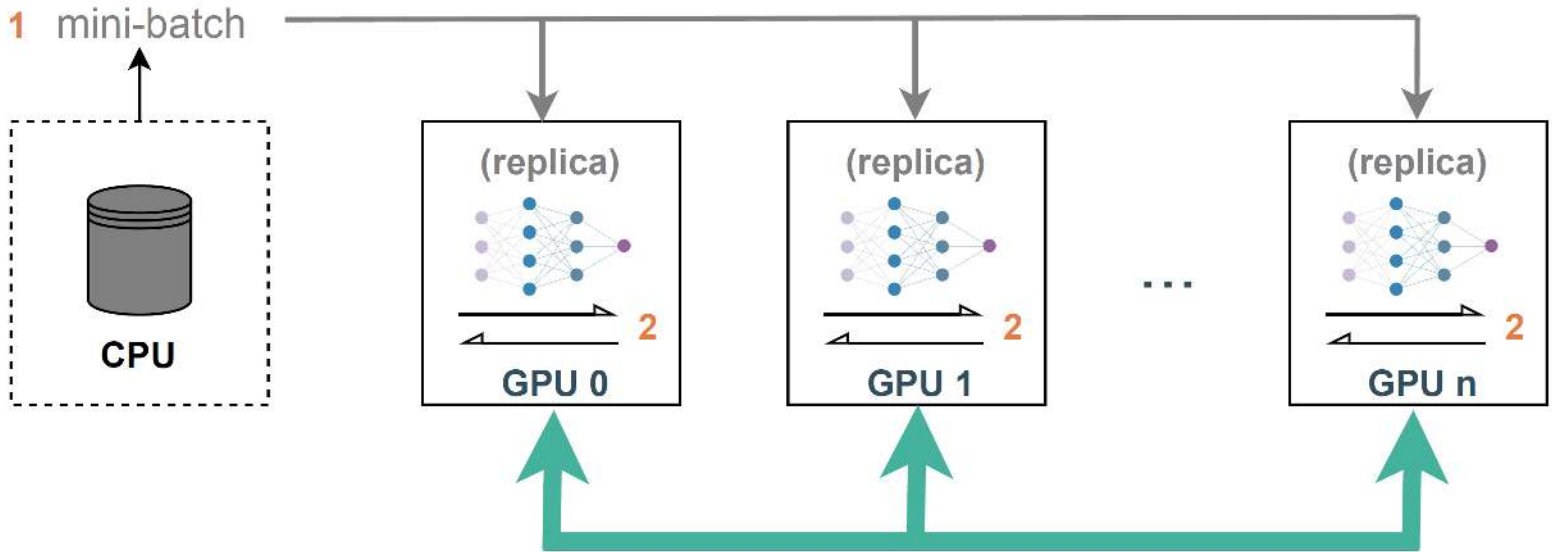
- Distributed DNN Training (Data parallelism)

Step 1: **Data Sharding** (training data is split into subsets distributed across GPUs).

Step 2: **FP + BP** (Each GPU holds an identical copy of DNN model parameters, and runs the forward and backward pass concurrently).

Step 3: **Gradient Aggregation** (Average the local gradients across all GPUs)

Step 4: **Parameters Update** with global gradients



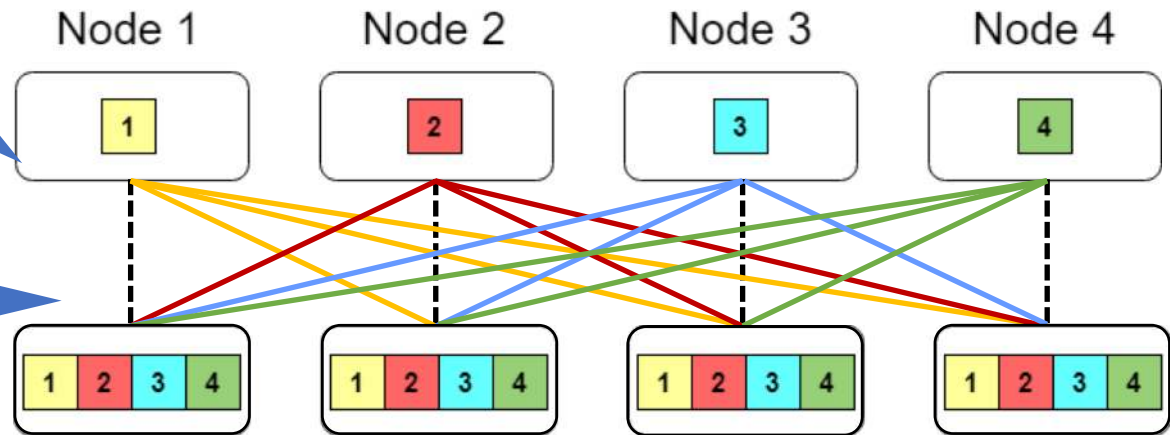
Expensive data movement
Gradient All-reduce

Photonic Communication

Step 3: **Gradient Aggregation**
(Average the gradients across GPUs)

Local gradients in each GPU node

Exchange gradients by inter-GPU
photonic communications



Photonic Communication

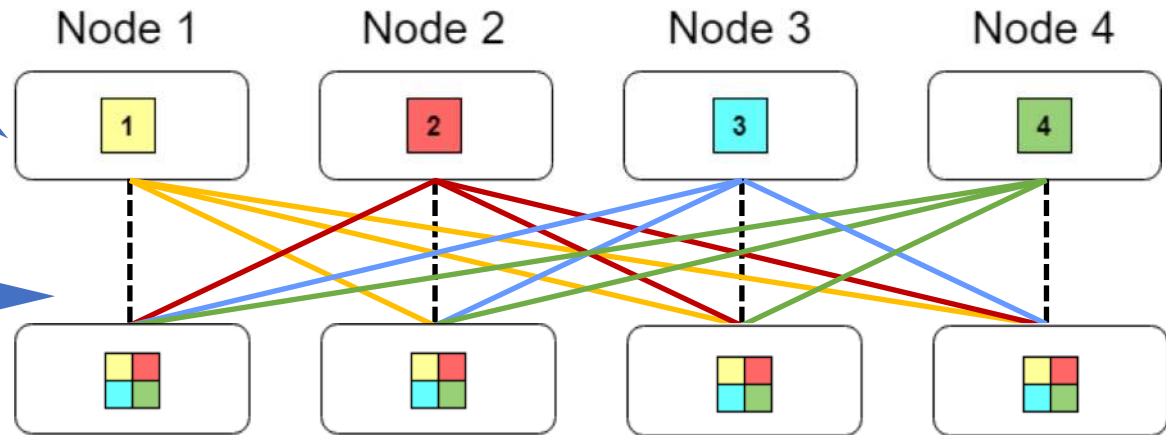
Step 3: **Gradient Aggregation**
(Average the gradients across GPUs)

Communication-intensive
Large number of gradients to transmit

Local gradients in each GPU node

Exchange gradients by inter-GPU
photonic communications

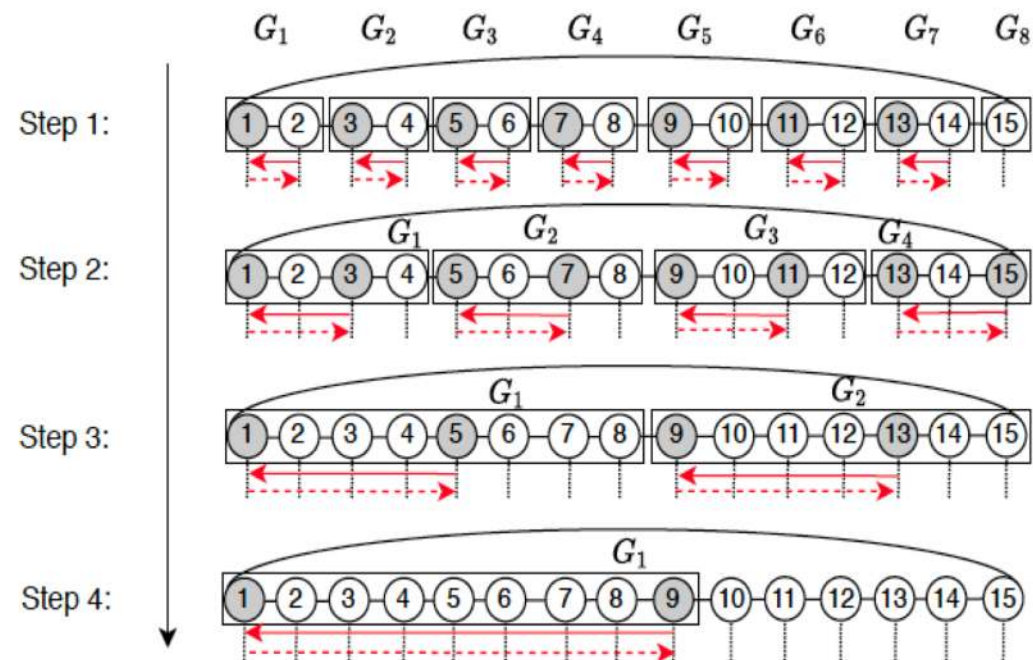
Average the gradients (global)



Photonic Communication

Motivation Example: Electronic Interconnect vs Photonic Interconnect

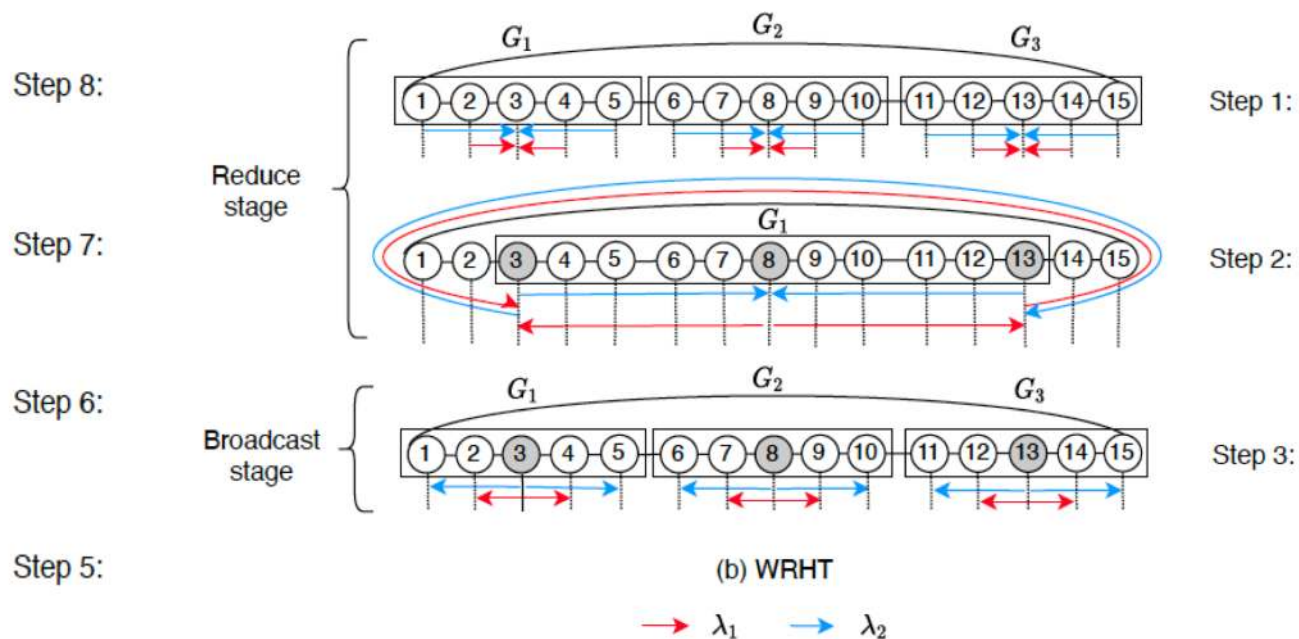
Reduce
Stage



(a) Binary tree all-reduce

Gradient All-reduce Algorithm
in Electronic Interconnect (8 steps)

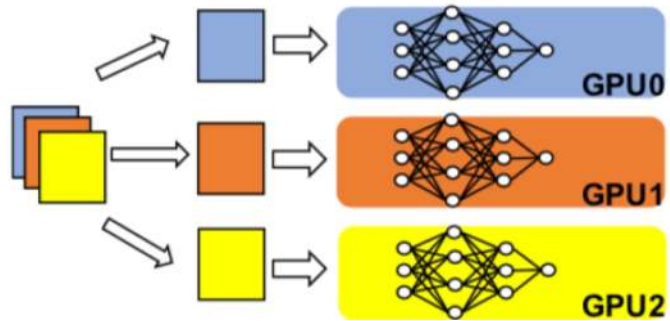
Broadcast
Stage



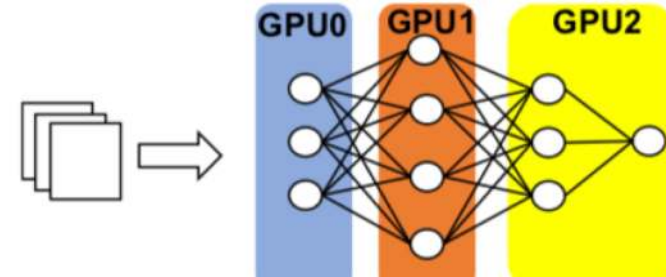
(b) WRHT

Gradient All-reduce
in Photonic Interconnect (3 steps)

Photonic Communication



Data Parallel



Model Parallel

Inter-GPU Photonic Interconnect

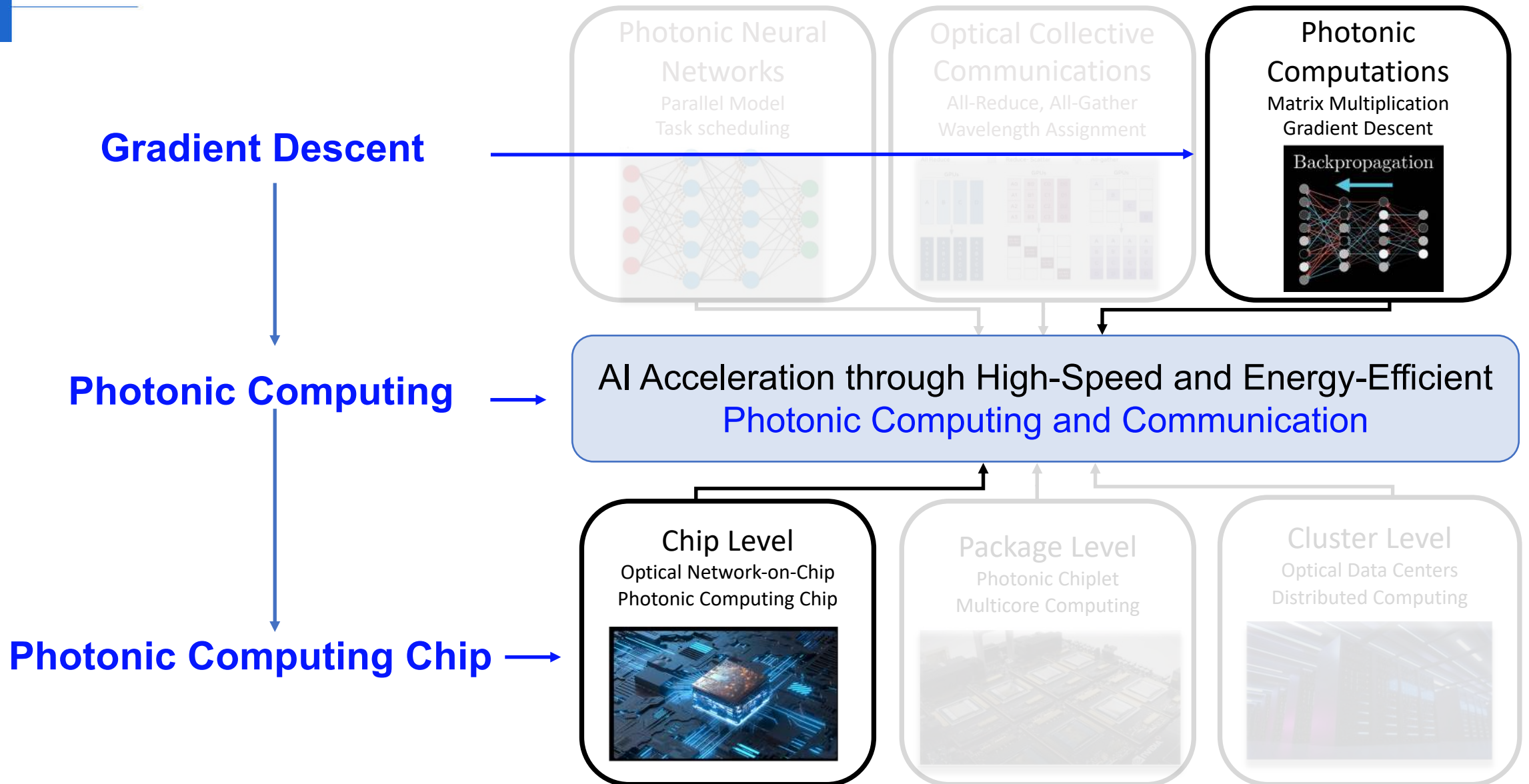
- [Data Parallel] WRHT: Efficient All-reduce for Distributed DNN in Optical Interconnect System, Fei Dai, Yawen Chen, et al, 52th **IEEE International Conference on Parallel Processing (ICPP)**, 2023.
- [Data Parallel] Efficient All-reduce for Distributed DNN in Optical Interconnect Systems, Fei Dai, Yawen Chen, et al, **ACM SIGPLAN Annual Symposium on Principles and Practice of Parallel Programming (PPoPP)**, 2023. **[Rank: A]**
- [Model Parallel] OpTree: An Efficient Algorithm for All-gather Operation in Optical Interconnect Systems, Fei Dai, Yawen Chen, et al, **IEEE International Conference on Communications (ICC)**, 2023. **[Flagship conference]**
- [Hybrid Parallel] Layer-wise Distributed DNN Training in Optical Interconnect Systems, Fei Dai, Yawen Chen, et al, in preparation to submit to IEEE Transactions on Computers.



SEECHIP

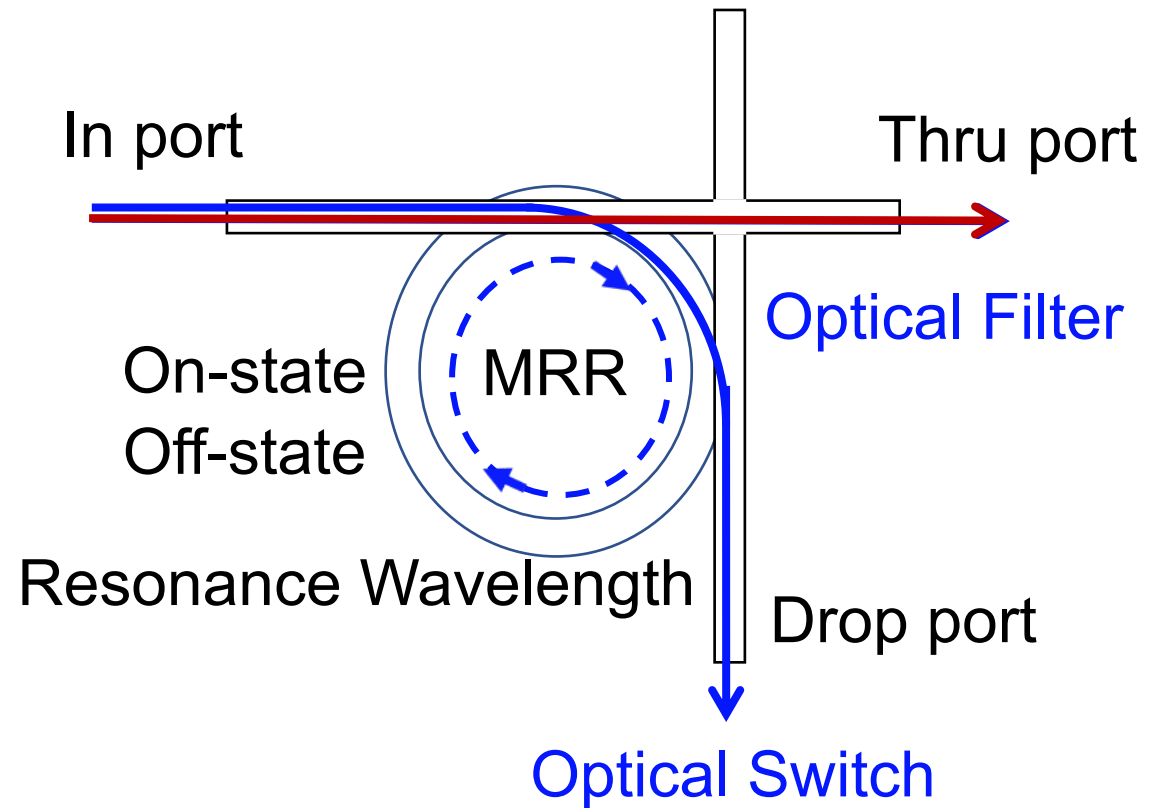
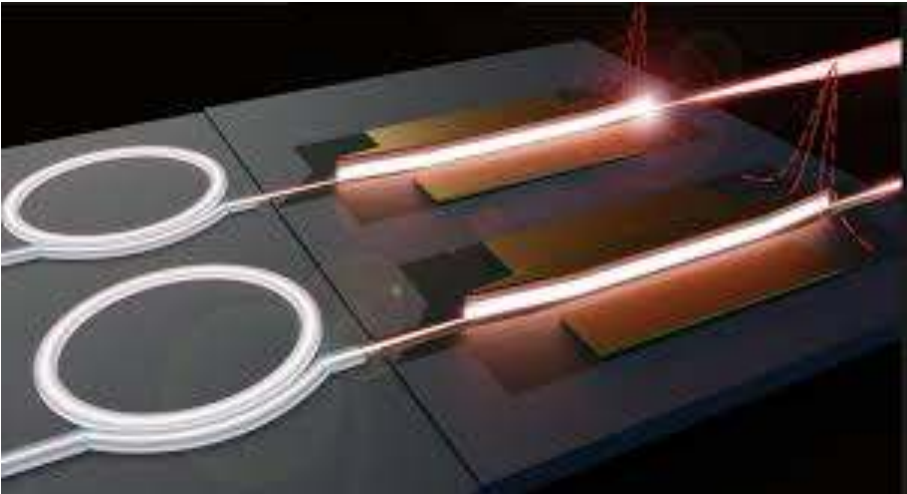
Hao Zhang, Yawen Chen, Zhiyi Huang, Haibo Zhang. SEECHIP: A Scalable and Energy-Efficient Chiplet-based GPU Architecture Using Photonic Link, International Conference on Parallel Processing (ICPP), 2023.

Photonic Computing



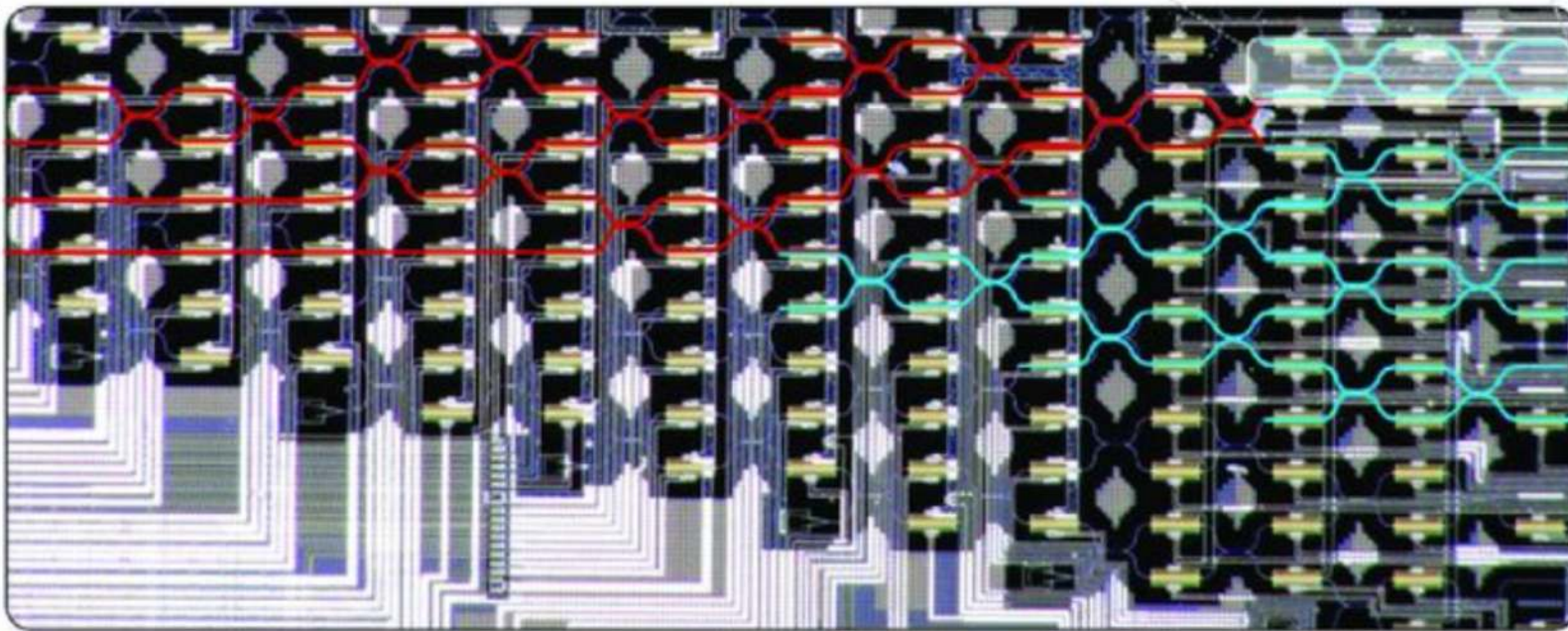
Photonic Computing (Key Components)

- Microring Resonator (MRR, Radius $\approx 5 \mu\text{m}$)
 - Radius $\approx 5 \mu\text{m}$
 - Optical switch, Optical filter

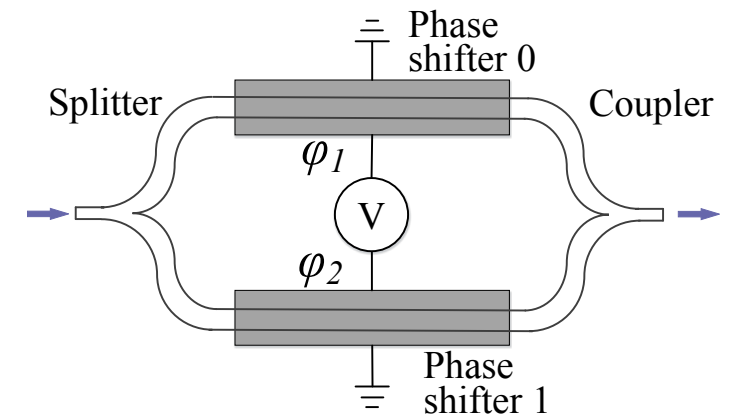


Photonic Computing (Key Components)

- Mach-Zehnder Interferometer (MZI)
 - Key optical component
 - Photonic Multiplication



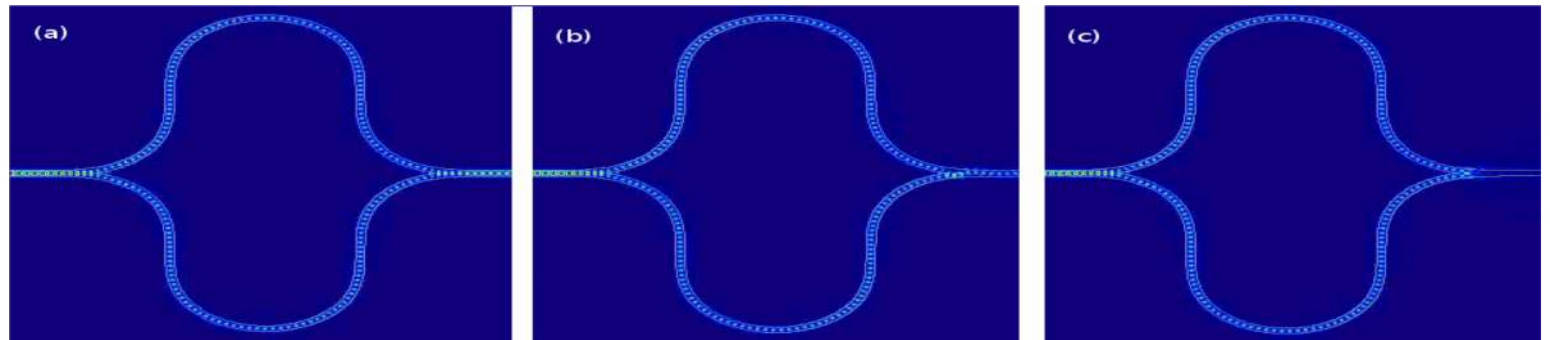
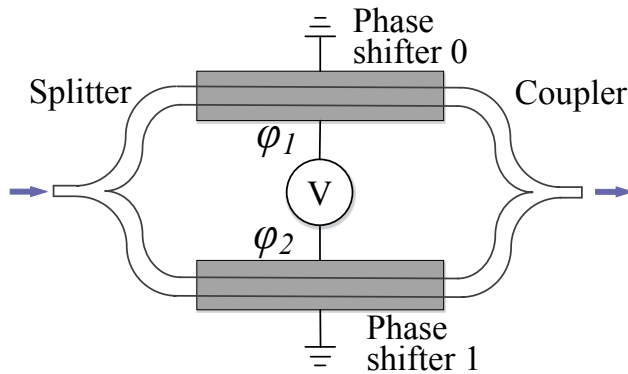
Shen & Harris, Nature Photonics 2017



Photonic Computing (Existing Devices)

- Photonic Multiplication with MZI

- Two different voltages are applied to the phase shifters in the two arms.
- $P_{out} = P_{in} \times \cos\left(\frac{\varphi_1 - \varphi_2}{2}\right)$.



$$\varphi_1 - \varphi_2 = 0$$

multiplied by 1

$$\varphi_1 - \varphi_2 = \pi/2$$

multiplied by 0.71

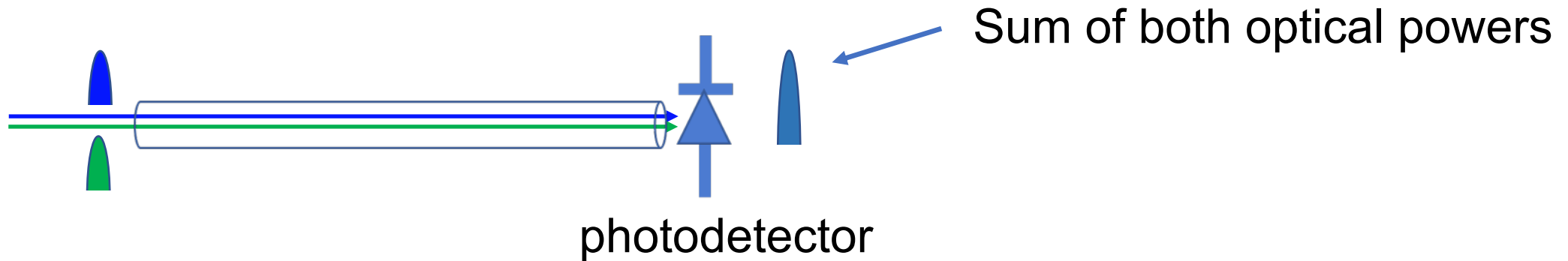
$$\varphi_1 - \varphi_2 = \pi$$

multiplied by 0

Photonic Computing (Existing Devices)

- **Photonic Addition**

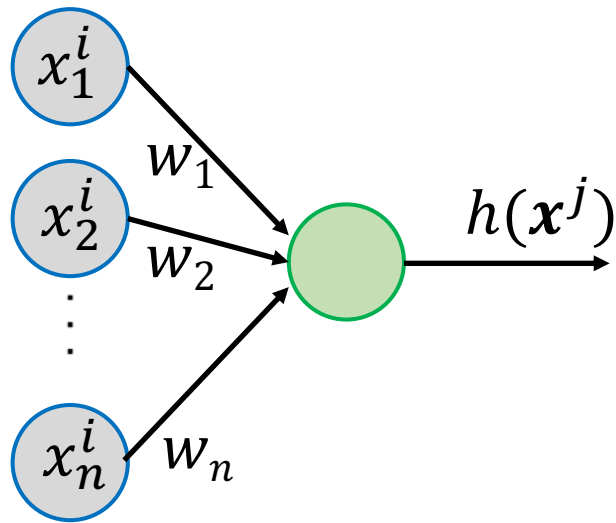
- Combine multiple optical signals in a single waveguide.
- Each optical signal is carried on a different wavelength.
- Photodetector outputs the sum of the all optical powers.



No power consumption is required for Photonic Addition!

Photonic MAC

- Existing photonic accelerators focus on accelerating forward propagation dominated by **Multiply and Accumulate (MAC)** operation

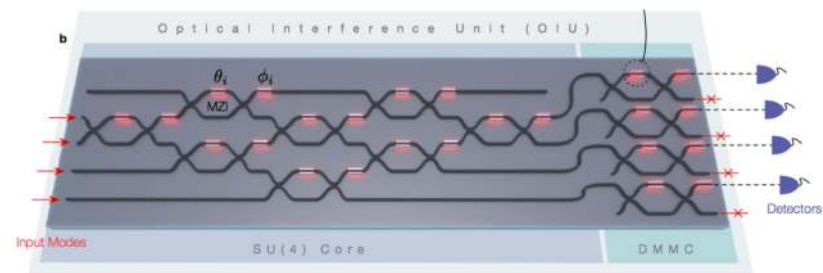


Activation function (e.g. ReLU) $\sigma(x) = \max(0, x)$

$$h(x^j) = \sigma \left(\sum_{j=1}^n w_j x_j^i + b \right)$$

← bias

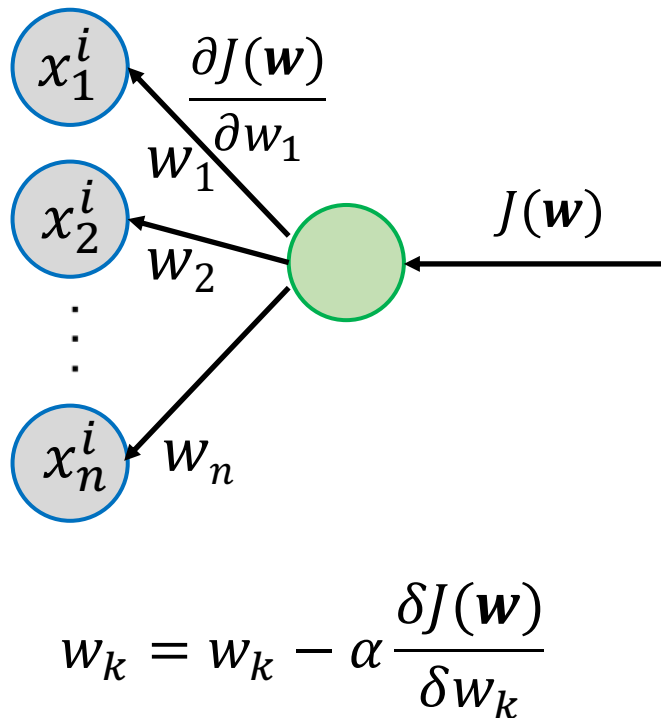
Multiply and Accumulate



Photonic Computing - STADIA

- **Problem:** Backpropagation is more computation intensive than forward propagation – gradient calculation for each weight

STADIA: Photonic Stochastic Gradient Descent for Neural Network Accelerators, Chengpeng Xia, Yawen Chen, Haibo Zhang, and Jigang Wu, **ACM Transactions on Embedded Computing Systems**, 2023.



Squared error loss: $J(\mathbf{w}) = \frac{1}{2} \sum_{i=1}^m (h(\mathbf{x}^i) - y^i)^2$

$$\frac{\partial J(\mathbf{w})}{\partial w_k} = \sum_{i=1}^m (h(\mathbf{x}^i) - y^i) \frac{\partial (h(\mathbf{x}^i) - y^i)}{\partial w_k}$$

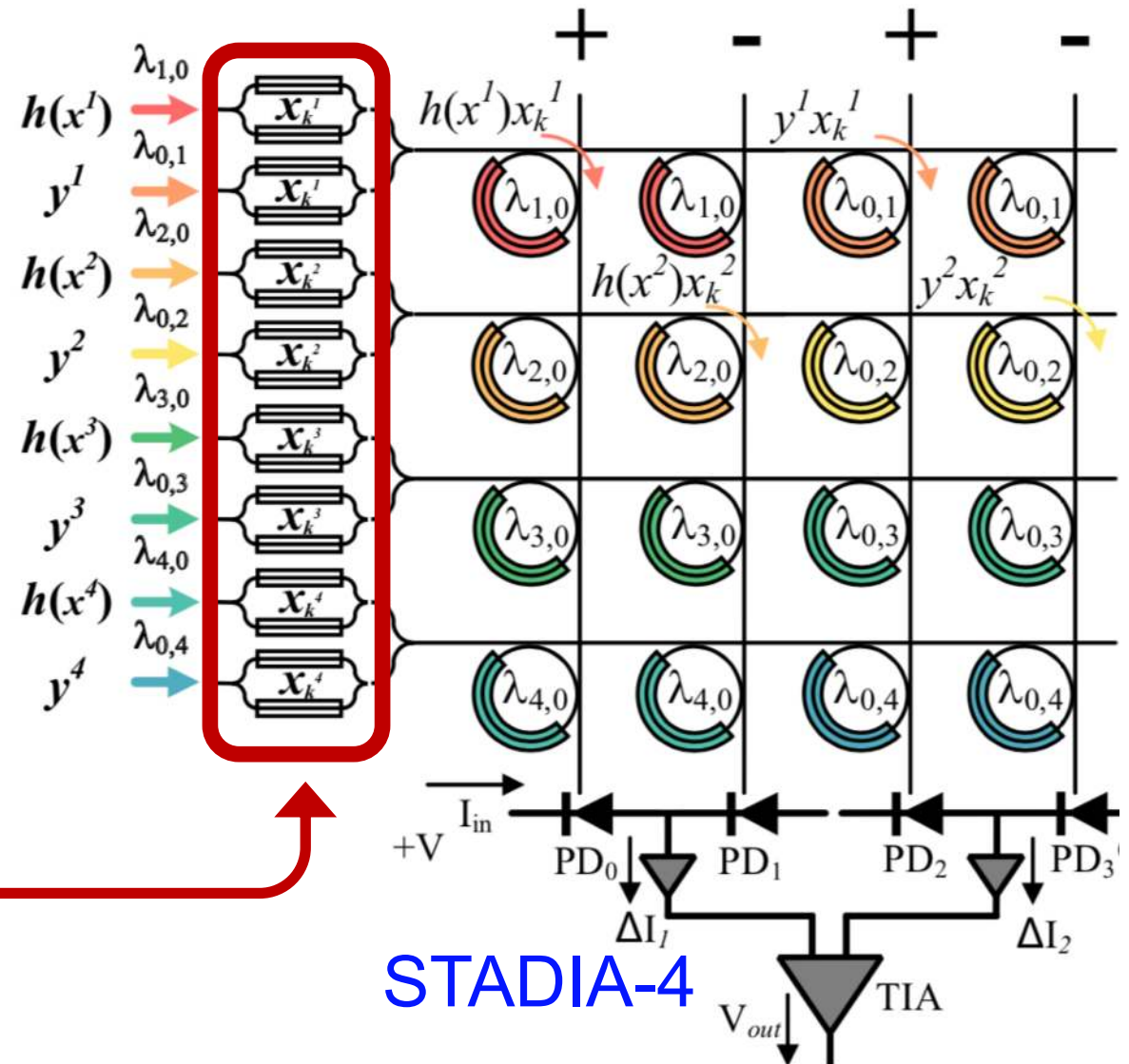
$$= \sum_{i=1}^m h(\mathbf{x}^i) x_k^i - \sum_{i=1}^m y^i x_k^i$$

Photonic Computing - STADIA

Photonic Gradient Descent

$$\frac{\partial J(\mathbf{w})}{\partial w_k} = \sum_{i=1}^m h(\mathbf{x}^i) \boxed{x_k^i} - \sum_{j=1}^m y^j \boxed{x_k^j}$$

Neuron input



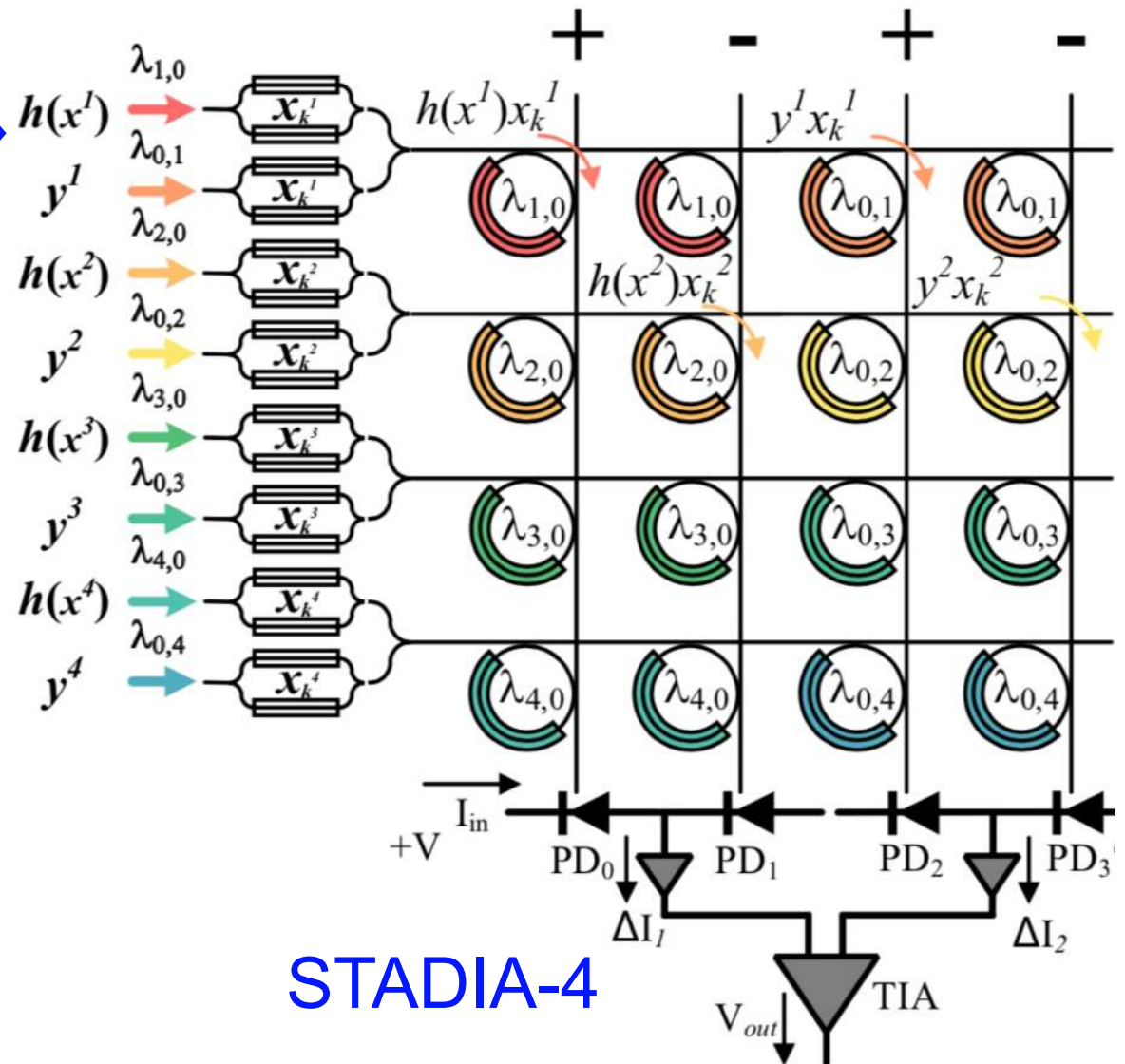
Photonic Computing - STADIA

Photonic Gradient Descent

$$\frac{\partial J(\mathbf{w})}{\partial w_k} = \sum_{i=1}^m \boxed{h(\mathbf{x}^i)} x_k^i - \sum_{j=1}^m \boxed{y^j} x_k^j$$

Prediction output

True label for sample i



STADIA-4

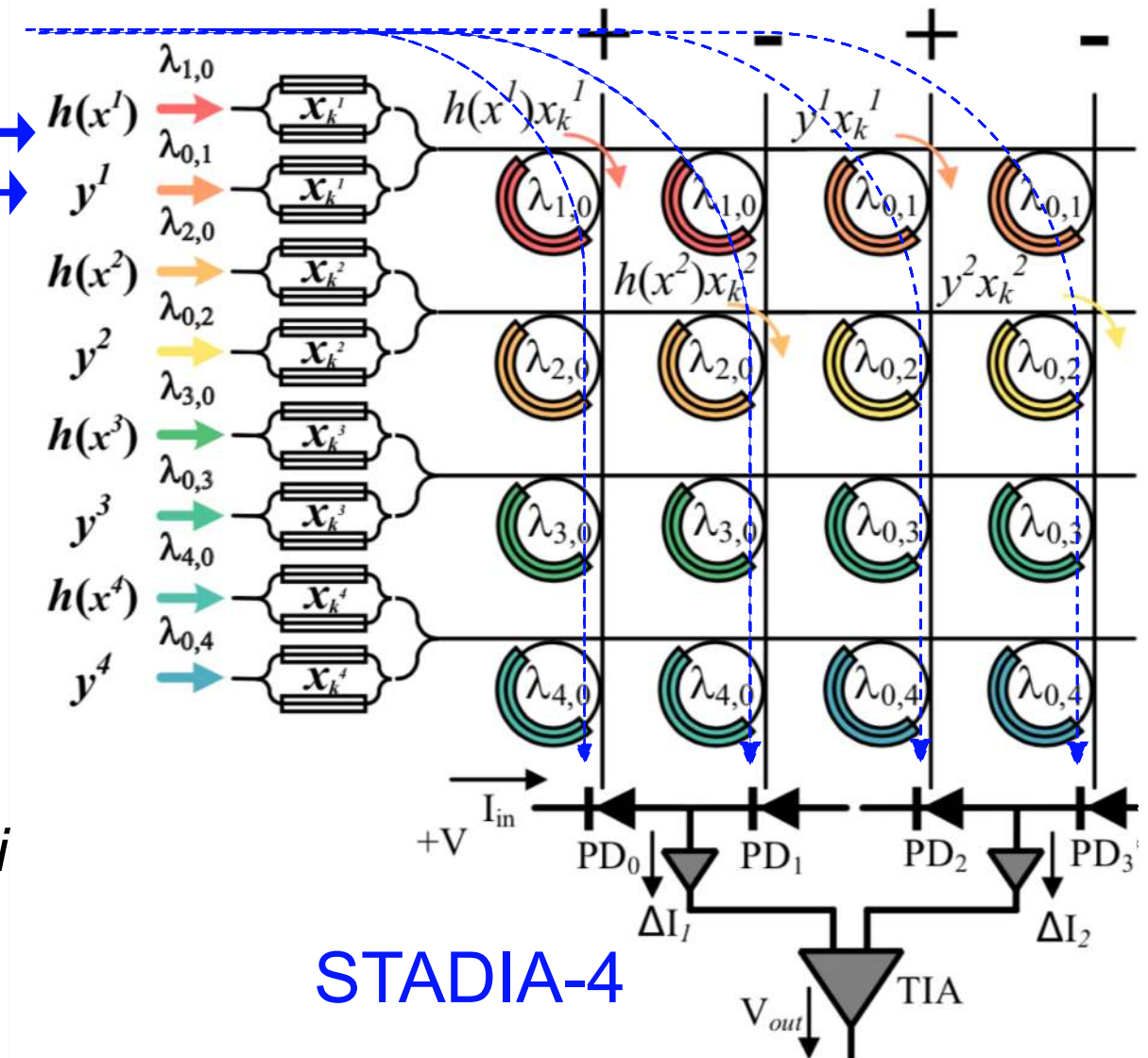
Photonic Computing - STADIA

Tuned to filter light signals

$$\frac{\partial J(\mathbf{w})}{\partial w_k} = \sum_{i=1}^m \boxed{h(x^i)} x_k^i - \sum_{j=1}^m \boxed{y^j} x_k^j$$

Prediction output

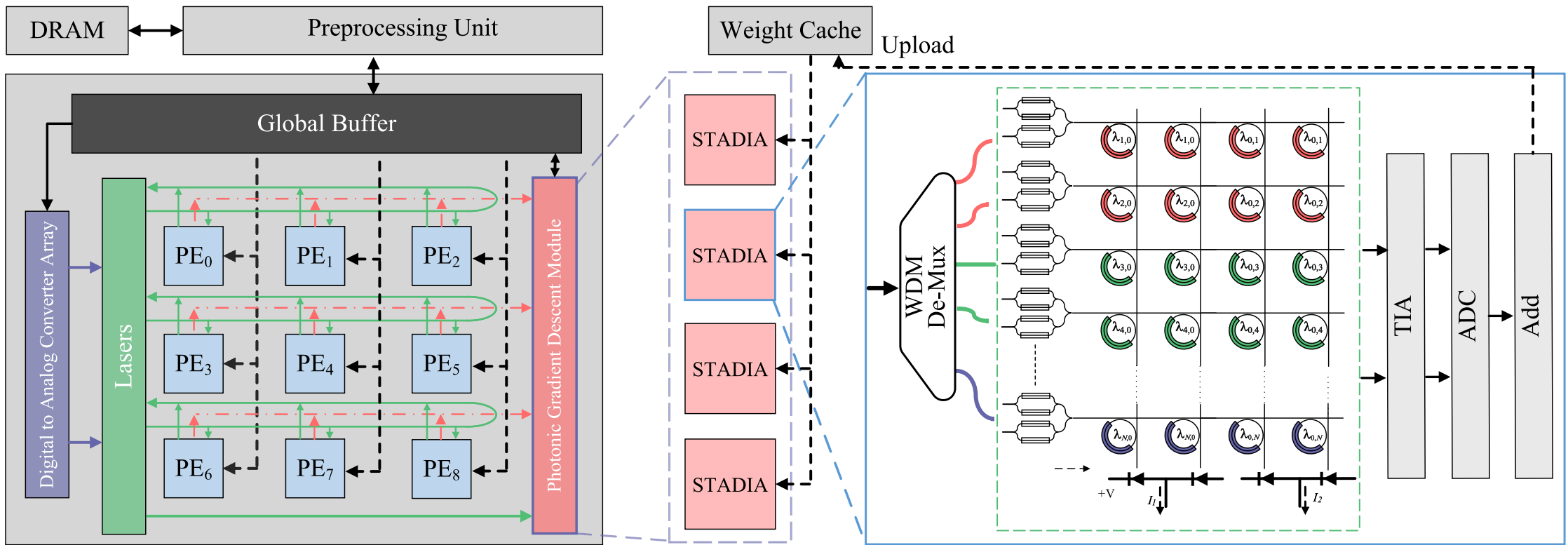
True label for sample i



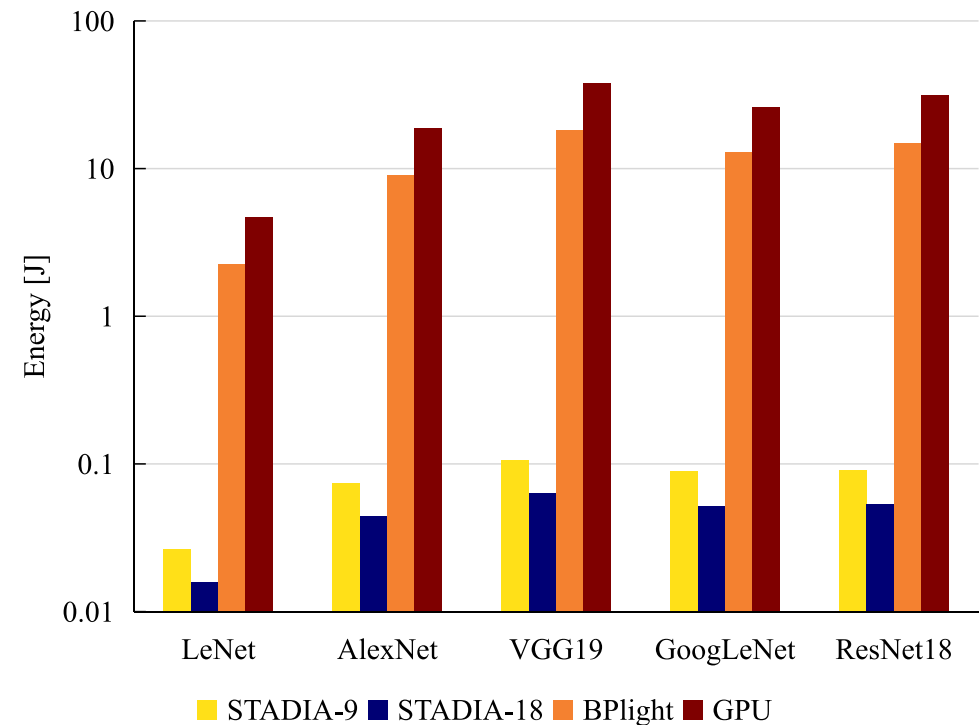
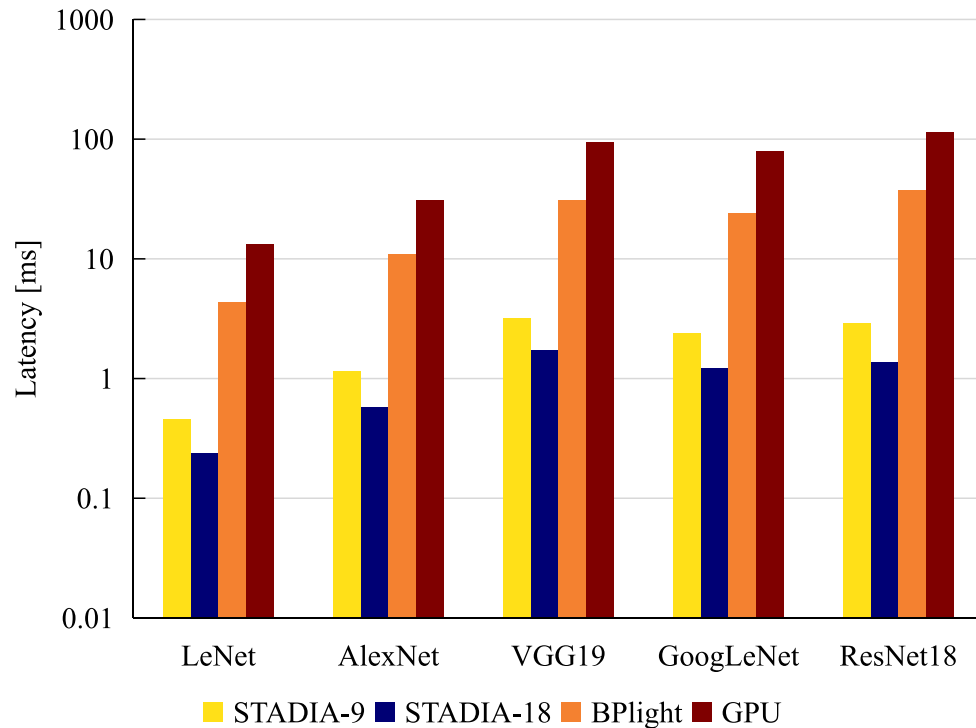
STADIA-4

Photonic Computing - STADIA

Photonic backpropagation architecture



Photonic Computing - STADIA

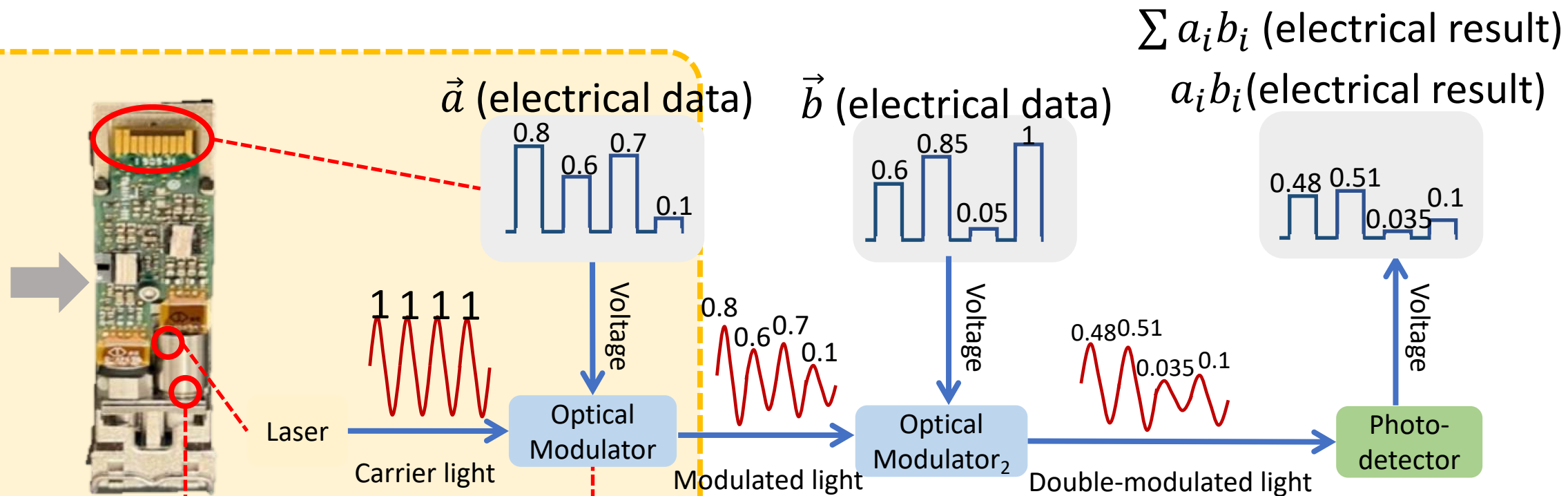


Compared with GPU, STADIA-9 and STADIA-18 **accelerate training by 31.8x and 63.3x**, while consuming **305.3x and 515.9x less energy**, respectively.

Photonic Computing - Lightning

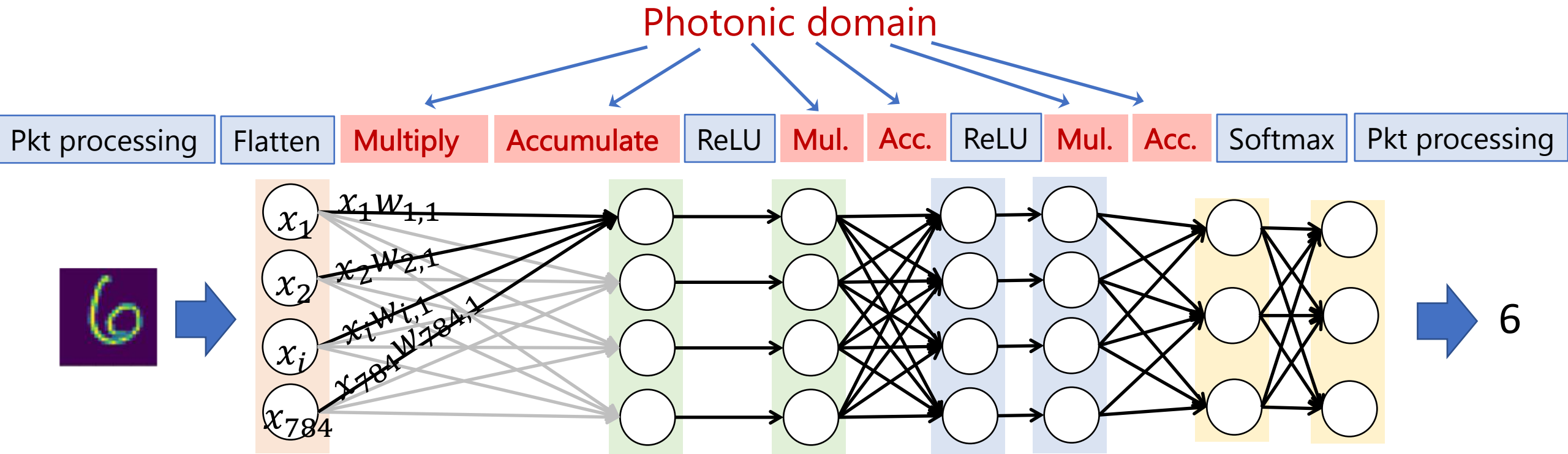
Zhizhen Zhong et al, Lightning: A Reconfigurable Photonic-Electronic SmartNIC for Fast and Energy-Efficient Inference, ACM SIGCOMM 2023.

Photonic computing: modulating light intensities as it propagates



Photonic Computing - Lightning

Challenge in AI acceleration: photonic operations are mixed with electronic ones

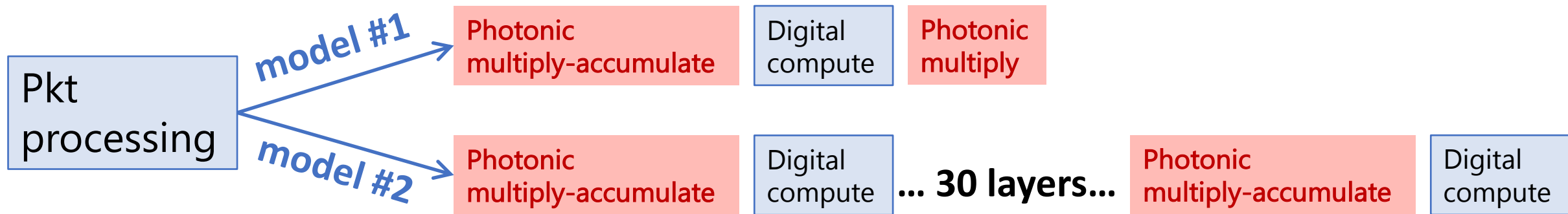


- A control logic is needed to coordinate the operations across electronics and photonics.

Photonic Computing - Lightning

Challenge in AI acceleration: reconfigure photonic-electronic operations for different ML models

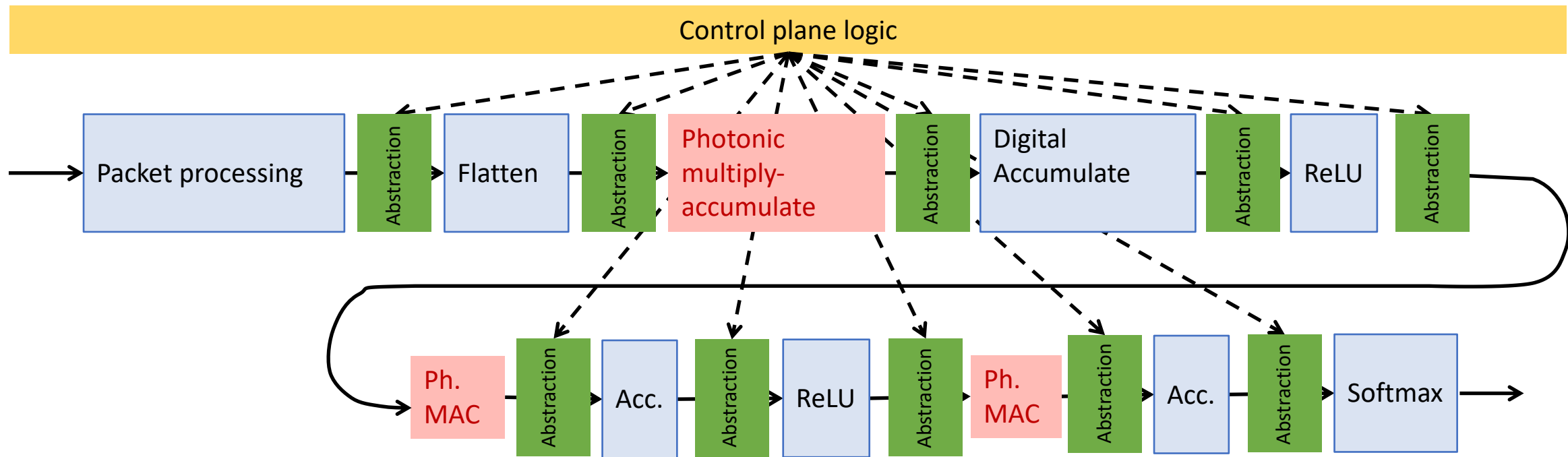
- Diverse ML inferences require a runtime reconfigurable system:



- ML models have relatively small number of computing “templates”: fully-connected layers, convolutions, ReLU, etc.
- Diverse number of combinations of “templates”: e.g., 30 convolution layers.

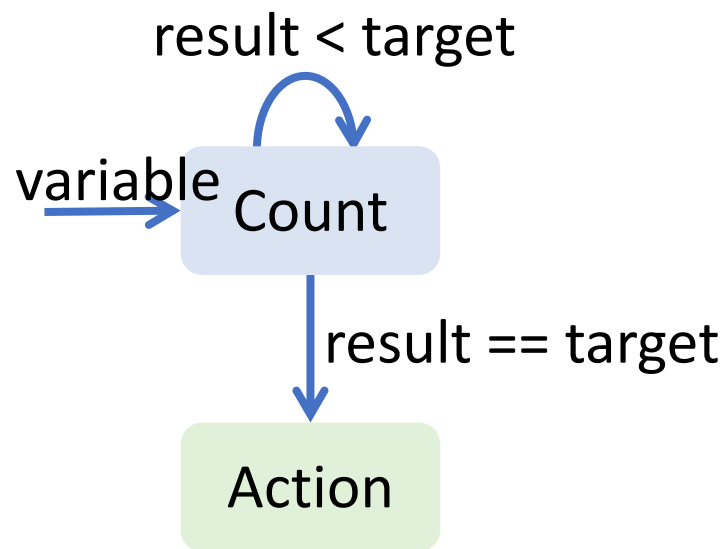
Photonic Computing - Lightning

Key idea: a hardware programming abstraction to make control decisions on the data plane.



Photonic Computing - Lightning

Lightning's reconfigurable count-action abstraction

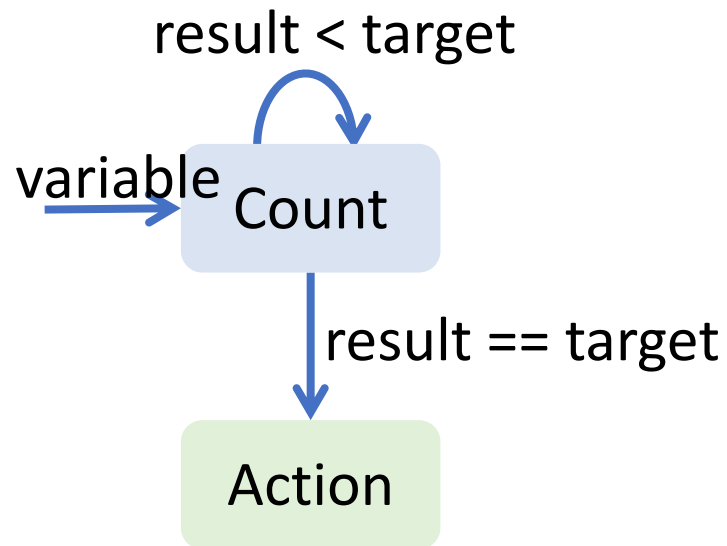


```
module datapath_module {  
    counts: {  
        // variables to be counted  
    }  
    targets: {  
        // a set of target results  
    }  
    actions: {  
        // actions to be triggered  
        // when the result is  
        // equivalent to the target  
    }  
};
```

High-level idea: Trigger an **action** whenever the **count** result reaches the **target**.

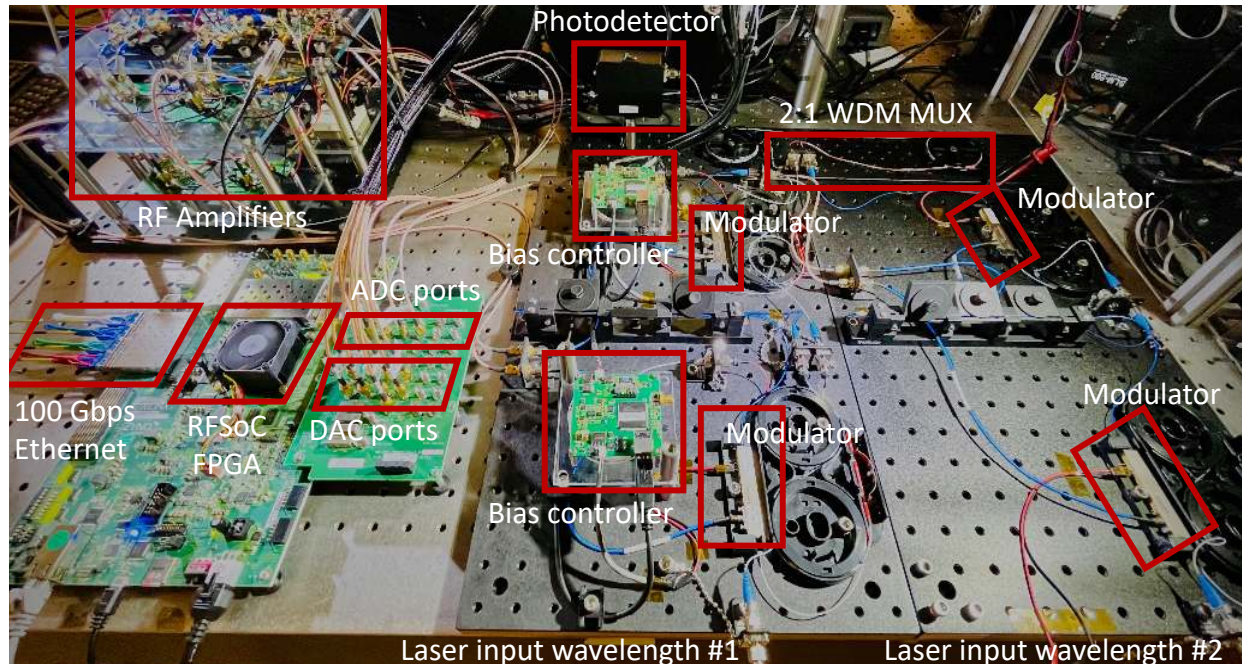
Photonic Computing - Lightning

Lightning's reconfigurable count-action abstraction

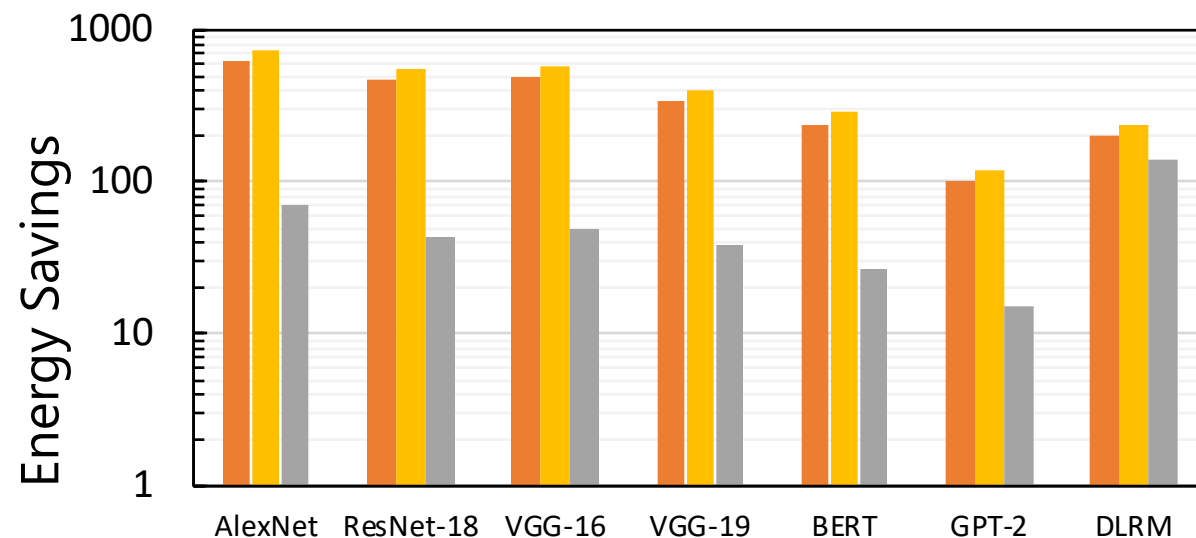
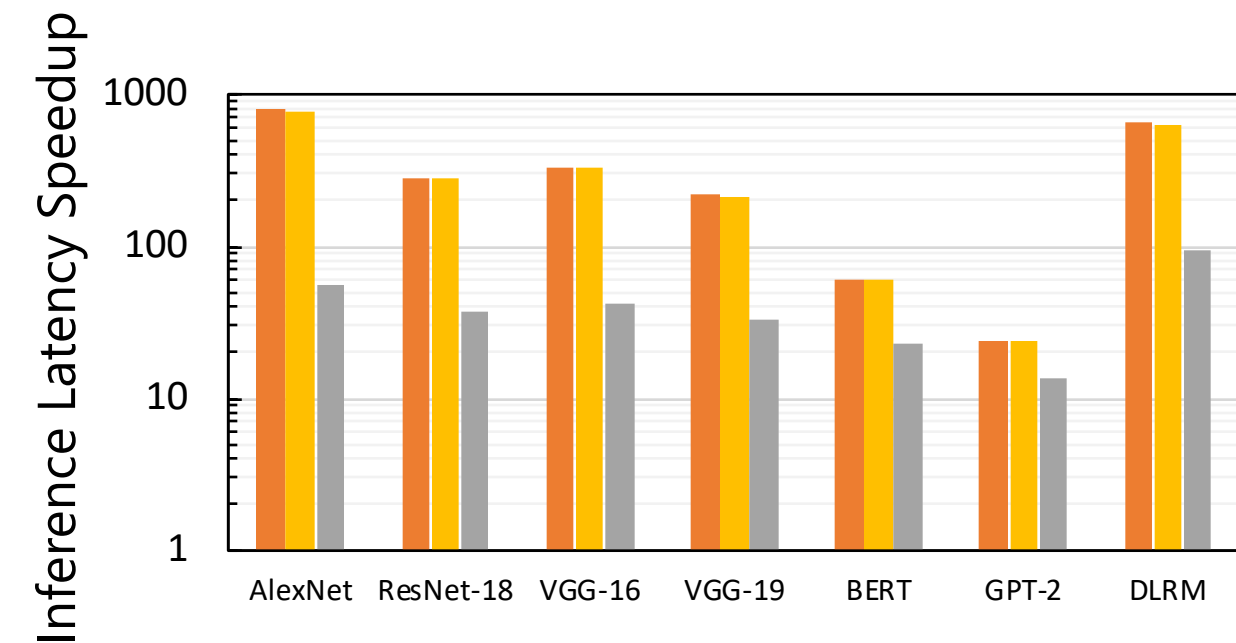


```
module synchornous_data_streamer {  
  counts: {  
    // count the sum of valid DAC flags  
    sum DAC[i].valid (i = 1 to num_DACs)  
  }  
  targets: {  
    // trigger when the sum equals the number of DACs  
    num_DACs  
  }  
  actions: {  
    // stream DACs' data into photonic cores  
    stream DAC[i].data (i = 1 to num_DACs)  
  }  
};
```

Lightning Testbed & Development Toolkit

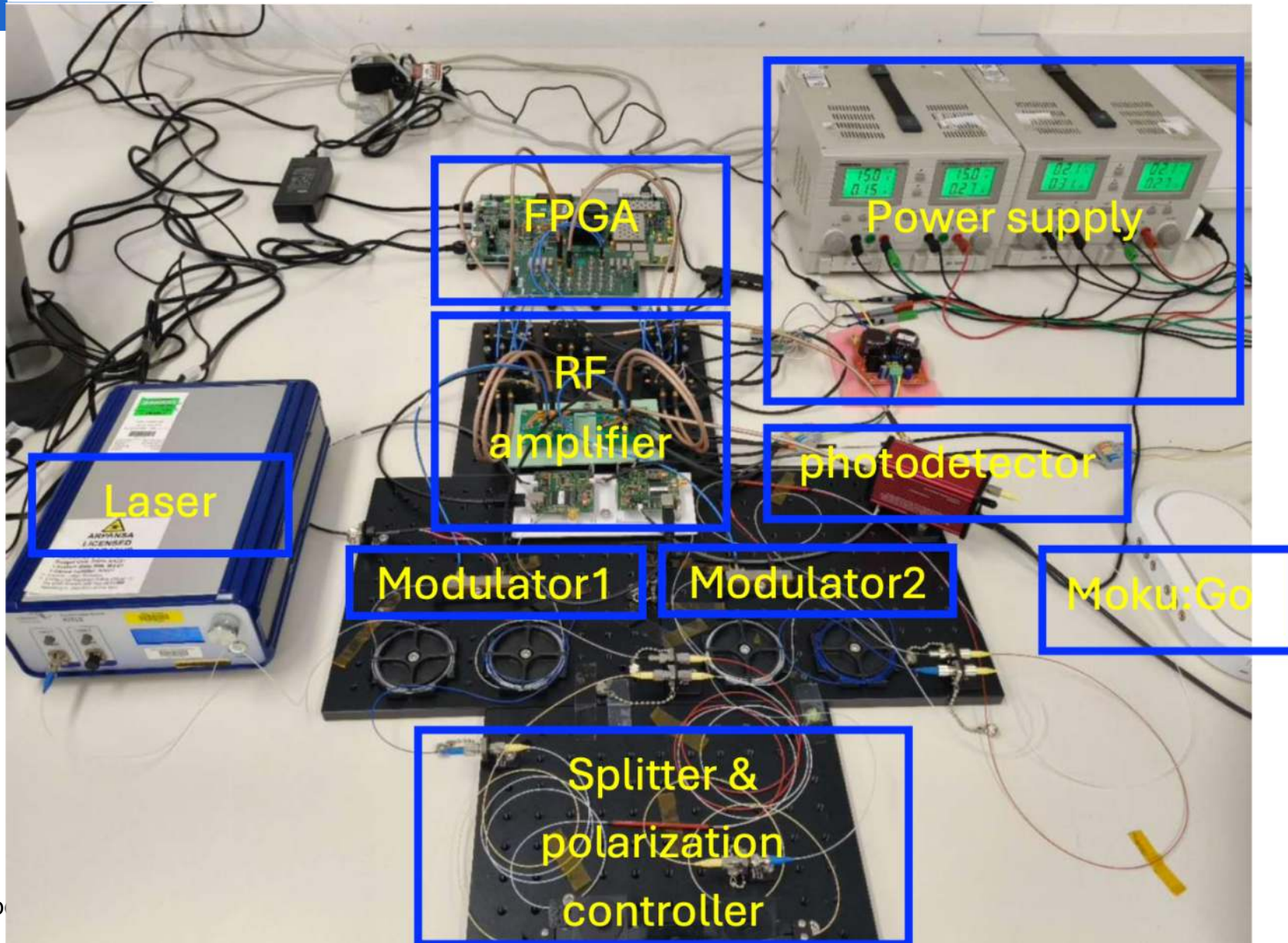


Lightning: latency and energy gains



Lightning improves the average inference serve time by $42\times$ - $337\times$ while reducing the energy consumption by $54\times$ - $419\times$.

Photonic computing testbed @ ANU



Photonic computing demo

Photonic computing: challenges

- Integration with Electronic Components
 - Electro-optic hybrid design due to the lack of optical memory
 - Low-loss and high-speed electro-optic conversion is essential to improve the performance of photonic computing
- Data Communication Bottlenecks
 - Data movement between photonic components and electronic memory can create bottlenecks
- High computation precision
 - Photonic devices can be sensitive to temperature variations
 - Photonic signals can experience attenuation as they travel through optical fibers or integrated photonic circuits.